



DR. NOUSHEEN MUSHTAQ

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DESIGNATION

Associate Professor
2011-till to date

Department of Pharmaceutical Chemistry
Faculty of Pharmacy, University of Karachi, Karachi

Assistant Professor
2004-2011

Department of Pharmaceutical Chemistry
Faculty of Pharmacy, University of Karachi, Karachi

Lecturer
2002-2004

Department of Pharmaceutical Chemistry
Faculty of Pharmacy, University of Karachi, Karachi

Cooperative teacher
(Full time)
2000-2002

Department of Pharmaceutical Chemistry
Faculty of Pharmacy, University of Karachi, Karachi

EDUCATION

Ph.D.
2004

Department Of Pharmaceutical Chemistry,
Faculty of Pharmacy, University of Karachi, Karachi.

M.Phil.
1999

Department Of Pharmaceutical Chemistry,
Faculty of Pharmacy, University of Karachi, Karachi.

Research Supervisor
M.Phil. & PhD.

Prof. Dr. Z.S. Saify (Former Dean Faculty of Pharmacy & Former Vice Chancellor University of Karachi). Presently at HEJ research institute of Chemistry (International center of chemical and biological Sciences) Karachi.

**AREA OF
INTEREST**

Medicinal Chemistry (drug designing, computational chemistry) Analytical Chemistry

TEACHING

Pharm-D, M.Phil. / M.Pharm. and PhD Classes
Organic Chemistry/Analytical Chemistry/ Medicinal Chemistry (Labs and Theory)

ON LINE COURSE
Nominated by HEC

LTTS Department, Indiana University USA

- 1- Integration of technology in to my class room** 2005
- 2- Integrating Technology into individual search projects** 2006
- 3- The information search process** 2006

COURSES DESIGNED

M.S. and **PhD.** related to drug design, research methodology and environmental chemistry

Pharm D. Lab (PHC-506) Preparation of Chemical Compounds identification and Quality Control

**MANUALS FOR
LABORATORY**

Designed Lab Manual for Lab PHC 506 comprising material on brief introduction and preparation of different chemical compounds, Lab techniques and instrumental methods for their identification

Manual for Lab PHC-607 based on Pharmaceutical Analysis (Chemical Methods)

**CATALOGUE
COMMITTEE**

Member catalogue committee representing department for revision and modification in courses

**TEACHERS TRAINING
PROGRAM**

Attended

Advanced Level Overseas Training Program at AIT Bangkok from Ensuring Training Effectiveness. (16th Jan. to 3rd Feb. 2012). Organized by Learning Innovation Department, Higher Education Commission in collaboration with Asian Institute of Technology (AIT) in Thailand.

Faculty development Course (19th July-15th Oct. 2004). Organized by Learning Innovation Department of Higher Education Commission.

Organized

Staff development program (May 28-2006-24Jun-2006). Organized by Human Resource Center University of Karachi.

Staff development program (Dec-2005-Jan-2006). Organized by Human Resource Center University of Karachi

EXTERNAL EXAMINER

- Hamdard University
- Baqai medical University
- Ziauddin Medical University,
- College of Pharmacy DowUniversity of Health sciences
- Jinnah Medical and Dental college Karachi

STUDENT ACTIVITIES

Patron of **Scientific Society of Pharmacy students**.

Started First Official News Letter of Faculty.
Pharma Times Jan 2009-till todate

RESEARCH

HEC approved supervisor (Higher Education Commission selects and approves research scholars throughout the country on the basis of research productivity and standard to supervised students who get HEC scholarship.

PhD. Students

1. Ms. Huma Naseem (Degree awarded in 2014)

Topic: *“Computer based drug designing of some nitrogen containing heterocyclic compounds for analgesic activity”*

2. Ms. Shazia Haider (Degree awarded in 2015)

Topic: *“Synthesis of some fluorine substituted piperidine derivatives and their biological evaluation”*

3. Ms. Rabya Munnawr (work in progress)

Topic: *“Acridine derivatives as antialzheimer agents; Insilico studies, synthesis and biological studies”*

4. Ms. Sumaira Ansari (work in progress)

Topic: *“Docking and synthesis of Indole and piperidine derivatives as neuro active compounds”*

M. Phil Students

- 1. Kashif Abdullah Shaikh (Degree awarded 2006)**
Topic: *"Synthesis, SAR & Biological Studies of Some N-substituted Phenacyl and Bromoethyl indole derivatives of Piperidine"*
- 2. Ms. Sumiyya Siddiqui (Degree awarded 2008)**
Topic: *"Synthesis & SAR of some novel analogues of 4-(4-Bromophenyl)-4-Piperidinol having potential biological activities"*
- 3. Sahazia Haider (Degree awarded 2010)**
Topic: *"Synthesis and SAR of some novel analogues of 4-(1-Pyrrolidinyl) Piperidine having Potential Biological Activities"*
- 4. Ms. Samreen Afzal (Degree awarded 2011)**
Topic: *"Synthesis and Pharmacological Studies of some Piperidine and Morpholine derivatives"*
- 5. Ms. Ayesha Akram (Degree awarded 2011)**
Topic: *"Synthesis of 4-(4-chlorophenyl)-4-hydroxy piperidine derivatives and their evaluation for biological activities"*
- 6. Ms. Rabya Munnawar (Degree awarded 2011)**
Topic: *"Synthesis and biological activities of Some 9-aminoacridine derivatives"*
- 7. Ms. Sadia Arif (Degree awarded 2012)**
Topic: *"Synthesis of 4-Aminomethyl Piperidine for analgesic and behavioral activities"*
- 8. Ms. Sana Anwar (Degree awarded 2013)**
Topic: *"Synthesis of novel aminomorpholine derivatives and their biological evaluation"*
- 9. Ms. Sadia Meer (Degree awarded in 2014)**
Topic: *"Synthesis of some new biological active derivatives of tryptamine"*
- 10. Mr. Jawad Ahmed (Degree awarded in 2015)**
Topic: *"Stability Analysis of Telmisartan Tablets Manufactured by Direct Compression Method"*
- 11. Ms. Noor un Namia (Thesis writing)**
Topic: *" Synthesis of some novel derivatives of 1-aminoanthraquinone and 4-aminoacetanilide and evaluation for anticancer antibacterial activity"*
- 12. Mr. Yasir Ahmed (work in progress)**
Topic: *"Computational studies of 9-amino acridine derivatives for antimalarial activity"*
- 13. Areesha Siddiqui (work in progress)**
Topic: *Docking, synthesis and analgesic activity of 4-amino acetanilide derivatives*

14. Saira Asghar (work in progress)

Topic: *Tryptamine derivatives as neuroactive and inflammatory agents. Computational studies, Synthesis and biological studies*

15. Ammara Iqbal (work in progress)

Topic: *Target based Synthesis of piperidine derivatives as psychotherapeutic agents*

M .Pharm. Students

1. Ms. Nazish Saba (Degree awarded 2010)

Topic: *"Synthesis of some new benzene sulfonyl derivatives of 4-(4-chlorophenyl) 4-hydroxypiperidine and their pharmacological evaluation"*

2. Ms. Munnawar (Degree awarded 2010)

Topic: *"Docking studies and synthesis of some azaindole derivatives for antimalarial activity"*

3. Ms. Faiza Khalil (Degree awarded 2011)

Topic: *"Validated spectrophotometric methods for the determination of Verapamil HCl in raw material of Verapamil hydrochloride"*

4. Ms. Nazia Zaheer (Degree awarded 2011)

Topic: *"Synthesis and SAR of 7-azaindole derivatives as potential acetyl choline esterase and butyrylcholine esterase inhibitors for Alzheimer disease"*

5. Ms. Gulnaz Sultana (Degree awarded 2011)

Topic: *"Synthesis of some morpholine derivatives having potential biological activity"*

6. Ms. Noureen Ashraf (Degree awarded 2011)

Topic: *"Synthesis of some bromoethylindole derivatives for analgesic and behavioral activity"*

7. Ms. Sumaira Ansari (Degree awarded 2011)

Topic: *"Synthesis of some derivatives of 4-piperidinopiperidine and their evaluation for potential therapeutic activity"*

8. Ms. Javeria Iqbal (Degree awarded)

Topic: *"Synthesis, docking and antimicrobial activity of 9 aminoacridine derivatives"*

9. Ms. Sadaf Ahmed (Writing thesis)

Topic: *"Synthesis and biological activities of indole derivatives"*

**SEMINARS /WORKSHOP
ORGANIZED**

1. **Research methodology and scientific paper writing skill** by Dr. Darakhshan J Haleem (HEJ) on April 29 2013
2. **Work Flow of Manufacturing and Analysis in Pharmaceutical Industry** by Mr. Muhammad Mushahid Riaz (Pfizer Lab. Pakistan) 8th September 2007
3. **Dimagh Ki Batti Jallao** by Mohammad Muneeb Kidwai (Corporate Training Manager Pfizer) 12th April 2008
4. **Breast Cancer** by Dr. Azmina Taj Valimohammad (Medical Oncologist Aga Khan University Hospital 9th August 2008
5. **Pharmaco-vigilance** by Dr. Muhammad Aslam Technical Director, Otsuka Pakistan Limited 30th August 2008
6. **Lymphoma Cancer and related Issues** by Dr. Salman Naseem Adil Head Hematology and Medical Oncology Aga Khan Hospital 1st November 2008
7. **Pharmaceutical Operations and Social Responsibilities of Pharmacist** by Mr. Mohammad Abdullah Feroz (M.D. Efroz Chemical Industries Ltd. 5th November 2008
8. **Zinc and its Efficacy in the treatment of Diarrhea** by Mr. Omar Taha Public Health Consultant Nutriset France 15th November 2008

**CONFERENCE
ORGANIZED**
Member Organizing
committee

2nd International Conference on Pharmaceutical Sciences, held in Karachi from 2nd to 5th Jan 1997.

CONFERENCES/ SEMINARS ATTENDED

1. Fifth International symposium –cum training courses on molecular medicine and drug research organized by Dr. Panjwani Center for Molecular Medicine and Drug research, International Center for Chemical and Biological Sciences held on 12-15 January 2015
2. Cancer an ongoing life threat; Seminar organized by department of physiology 2014
3. 14th Asian Symposium on Medicinal Plants, Spices and other Natural Products organized by HEJ Research Institute of Chemistry, International Centre for Chemical and Biological Sciences, Karachi. Dec. 9-12, 2013
4. Fat diet and healthy weight management by Dr. Capt. Adnan Saleem organized by seminar committee on 25th March 2013
5. Improve your performance through mind-body relaxation organized by seminar committee on 21st March 2013
6. Workshop on Intellectual Property Rights, organized by KU-ORIC and PASTIC at KU. June 2, 2012
7. Pharmacist: A need of health care system, symposium organized by Ziauddin College of Pharmacy. April 28, 2012
8. Workshop on Bioethics, Faculty of Pharmacy, University of Karachi. February 23, 2012
9. Cancer; an ongoing life threat, organized by Department of Physiology, University of Karachi. July 26, 2012.
10. Training of Trainers Workshop, Faculty of Pharmacy, University of Karachi. June 13-14, 2011
11. Clinical Pharmacy Practice. Symposium organized by Ziauddin College of Pharmacy, Ziauddin University. April 23, 2011
12. Workshop on Bioethics organized by Faculty of pharmacy held on 23rd February 2012
13. 3rd International Symposium cum training courses on molecular medicine and drug research organized by Dr. Panjwani Center for Molecular Medicine and Drug research, International Center for Chemical and Biological Sciences held on 3-6 January 2011
14. Symposium On Clinical Pharmacy practice organized by Ziauddin University held on 23rd April 2011
15. Seminar on Cancer an ongoing life threat organized by department of physiology 26th July 2011
16. Seminar on Fundamentals of GMP organized by In-group information Services on 23rd February 2010
17. Seminar on Hospital Pharmacy organized by Patel Hospital 2nd December 2009
18. Symposium on Hospital Pharmacy Practice organized by Ziauddin college of Pharmacy, Ziauddin University 17th November 2009
19. 11th International Symposium on Natural Product Chemistry, organized by International Center for Chemical and Biological Sciences (H.E.J Research Institute of Chemistry Dr. Panjwani Center for Molecular Medicine and Drug research) University of Karachi Pakistan 29th Oct.-1st November 2008.
20. International workshop on The European Directorate for the Quality Medicines (EDQM) held in Karachi, Jan 26-27, 2006.
21. Real Time PCR Workshop Department of Genetics University of Karachi, Karachi, 15th of August 2005.

22. One Day Seminar on Examination Standards organized by Quality Enhancement Cell and Department of Physiology, University of Karachi, Karachi, Pakistan-August 6, 2005
23. 15 AFASSA Symposium on Natural Products and regional Meeting, University of Karachi, July 25-27, 2005, Karachi, Pakistan.
24. 8th. Biennial National Conference of P.S.B.M. B. on “Emerging Trends in Biochemistry and Molecular Biology” University of Karachi, Pakistan – March 7-9, 2005.
25. 4th International symposium on “Emerging dimensions of health care” organized by Faculty of Health and Medical Sciences, Faculty of Eastern Medicine and Faculty of Pharmacy, Hamdard University, held in Karachi on 20-21st Jan. 2004.
26. 9th International Symposium on Natural Products, organized by HEJ Research Institute of Chemistry, International Center for Chemical Sciences, University of Karachi, held in Karachi on 9-13 Jan. 2004
27. Research Enhancement Workshop 2nd January, 2003 Organized by Higher Education Commission
28. 7th Eurasia Conference on Chemical Sciences, Held in Karachi, March 09-12, 2002.
29. Symposium on Untapped Fisheries Resources of Pakistan, Held in Karachi from Feb. 23-24, 2001.
30. Recent Advances in Psychopharmacology and its impact on the Management of Psychiatric Illnesses, held in Karachi from 19th Feb.- 1st March, 2001.
31. Workshop on “ PCR/DNA Fingerprinting for Forensic Investigations and Restriction Enzyme Digestion for cloning” held in Karachi from 10-11 July 2000
32. Seminar on Good Manufacturing Practices Quality Control Quality Assurance and Manufacturing Technology held in Karachi from 8th to 17th May 1998.
33. International Conference on “Focus on Pharmaceuticals National / International Issues: The Next Millennium” held in Karachi from 4th to 7th August 1997.
34. Workshop on the use of Bioassay Techniques for New Drug Development organized by HEJ Research Institute of Chemistry on July 8th and 9th, 1997
35. First one day seminar on Herb Medicine and Therapeutics held in Karachi on March 29th, 1997
36. Second ISBBP Symposium on Biochemistry and Biophysics held in Karachi on 5th and 6th March 1997
37. 2nd International Conference on Pharmaceutical Sciences, held in Karachi from 2nd to 5th Jan 1997.

RESEARCH WORK PRESENTED

1. **Synthesis and Antileishmanial Activity of Fluorine containing heterocyclic compounds.** Shazia Haider, Zafar Saied Saify, Mehrun Nisa, Nousheen Mushtaq, Shama Perveen, Tabinda Zareen, Nighat Sultana, Kiran Rafiq, Sadia Siddiq and M.I.Choudhary. 13th International Symposium on Natural Product Chemistry by HEJ Research Institute of Chemistry, International Center for Chemical and biological Sciences, University of Karachi, held in Karachi on September 22-25, 2012, Karachi, Pakistan. page no. 207
2. **Synthesis and biological Assessment of some piperidine, 7-Azaindol and benzimidazole derivatives as potential DHFR inhibitors.** Shazia Haider, Zafar Saied Saify, Pervez Iqbal, Afshan Naz, Mehrun Nisa, Arshad Aryne, Tabinda Zareen, Nousheen Mushtaq. 3rd

International Symposium –cum-Training Course on Molecular Medicine and Drug Research. January 3-6, 2011, 43

3. **Antiglycation Activity of 7-Azaindole derivatives.** Tabinda Zareen Zafar Saied Saify, Shazia Haider, Saima Rasheed, Mehru-Nisa, Pervez Iqbal, Afshan Naz, Munawer Khanum, Arshad Aryne, Kiran Rafiq, Nousheen Mushtaq, M.I.Choudhary. 3rd International Symposium –cum-Training Course on Molecular Medicine and Drug Research. January 3-6, 2011, p-44
4. **Synthesis Of Some 7-Azaindole Derivatives As Potential Urease Inhibitor** Zafar Saied Saify, Mehru-Nisa, Nousheen Mushtaq, Shazia Haider, Ajmal Khan, Tabinda Zarreen, Munawer Khanum, Sadia Siddiq and M.I.Choudhary. 12th International Symposium on Natural Products Chemistry, 22nd -25th December, 2010. pp-147
5. **Pyrrolidinyl-Piperidine Derivatives As CNS Activator.** Zafar Saied Saify, Nousheen Mushtaq, Mehru Nisa, Kaniz Fizza Azhar, D.J. Haleem, Shamim Akhtar & Shazia Haider. 12th International Symposium on Natural Products Chemistry, 22nd -25th December, 2010. pp-119
6. **Synthesis of Some Phenacyl based New Isonipecotamide derivatives as Analgesic and Antimicrobial agents.** Z. S. Saify, Khalid M. Khan, Shamim Akhtar, M. Arif, Nousheen Mushtaq, Mehru Nisa and Shazia Haider. 11th International Symposium on Natural Product Chemistry, October 29 –November 01, 2008, pp-146.
7. **Neurochemical studies of some Pyrrolidino-Piperidine Derivatives.** Z. S. Saify, Nousheen Mushtaq, Mehru Nisa, D.J. Haleem, Shamim Akhtar, and Shazia Haider. Proceeding of 11th International Symposium on Natural Product Chemistry, October 29 –November 01, 2008, pp-147.
8. **Synthesis of some Pyrrolidino Piperidine derivatives having potential Analgesic Activity.** Z. S. Saify, Nousheen Mushtaq, Shamim Akhtar, Mehru Nisa and Shazia Haider. 11th International Symposium on Natural Product Chemistry, October 29 –November 01, 2008, pp-148
9. **Molecular Approaches Towards Alzheimer's disease,** Faiza Tahir, Atta-ur-Rahman, M. Iqbal Choudhary, Zafar Saied Saify, Mehru Nisa, Shazia Haider and Nousheen Mushtaq., International Conference on Alzheimer Disease, April 2008, Chicago, U.S.A.
10. **Synthesis, Antioxidant and cytotoxic activity of some novel analogues of piperidine** Z.S. Saify, Nousheen Mushtaq, Kashif Abdullah Shaikh, Shamim Akhtar, M. Arif, Viqar-uddin Ahmed and M. Iqbal Choudry; 8th Biennial National Conference of Pakistan Society for Biochemistry and Molecular Biology 7th March 2005
11. **Synthesis and Antioxidant activity of some novel analogues of Nitrogen containing heterocyclic compounds;** Z.S. Saify, Nousheen Mushtaq, Shamim Akhtar, Viqar-uddin Ahmed, M. Iqbal Choudry, Faryal Mushatq, talat Makhmoor and Summaiya; 3rd National Conference on Preclinical Research on Product of biological origin in Hospitals; Pakistan Society of Pharmacognosy, 7th April 2004
12. **Piperidine-4-carboxamide and its congeners as potential Analgesic agents;** Z.S. Saify, Shamim Akhtar, M. Arif, Nousheen Mushatq, Rafiq Alam Khan and Sana; 3rd National

Conference on Preclinical Research on Product of biological origin in Hospitals; Pakistan Society of Pharmacognosy, 7th April 2004

13. **Synthesis, Antioxidant and Cytotoxic activity of some novel analogues of piperidine and pyrrolidine;** Z.S. Saify, Nousheen Mushtaq, Shamim Akhtar Viqar-uddin Ahmed, M. Iqbal Choudhary Faryal Mushtaq and Talat Makhmoor; 9th International Symposium on Natural Products held on Jan. 2004)
14. **Analgesic activity of some phenacyl derivatives of piperidine and pyrrolidine** Z.S. Saify, Nousheen Mushtaq, and Shamim Akhtar; 9th International Symposium on Natural Products was held on Jan. 2004.
15. **Synthesis of some isonipecotamide based phenacy derivatives as Anti-inflammatory agents** Z. S. Saify, Shamim Akhtar, M. Arif, Shaheen A. Husain, Nousheeen Mushtaq, Khalid M. Khan and Shahnaz Ahmed, 9th International Symposium on Natural Products was held on Jan. 2004.
16. **Synthesis and SAR Studies Of Piperidin-4-Carboxamide Derivatives Having Analgesic, Anti-inflammatory And Anti-microbial Activities** Z. S. Saify, Shamim Akhtar, M. Arif, Shaheen A. Husain, Nousheeen Mushtaq, Khalid M. Khan and Shahnaz Ahmed; 9th International Symposium on Natural Products was held on Jan. 2004
17. **CNS and Antibacterial activity of synthesized derivatives of carboline** Z. S. Saify, Waqar Hussain, Javed Farhad, Nousheen Mushtaq, Fozia Noor, Shamoona Takween Shamim Akhtar, Khalid M. Khan and Farah Mazhar, 9th International Symposium on Natural Products was held on Jan. 2004.
18. **Some Azaindole derivatives of potential Biological activities** Z.S. Saify, K.M. Khan, Nousheen Mushtaq, Fozia Noor, S. Tassadaque Ali Shah, Shamim Akhtar, and M.Arif; 8th International Symposium on Natural Product Chemistry; Jan18-22 2000, Karachi Pakistan.
19. **Effect of *Lachnanthes tinctoria* on certain behavioral parameters**, Z.S. Saify, Fozia Noor, Nousheen Mushtaq, Shamoona Takween and Shahida Ahmed, Proceeding of First one-day seminar on Herb, Medicine and Therapeutics held on March 29th, 1997.
20. **Biological Studies of herbal medicine *Passiflora incarnata***, Z.S. Saify, Fozia Noor, Nousheen Mushtaq, M. Arif, and Ahsana Dar, Proceeding of First one-day seminar on Herb Medicine and Therapeutics held on March 29th, 1997.
21. **Effect of ethanolic extract of *Bryonia dioica* on the spontaneous activity of isolated rabbit jejunum**, Z.S. Saify, Nousheen Mushtaq, Fozia Noor, M. Arif, Shehnaz Naim and Ahsana Dar, (Proceeding of First one-day seminar on Herb, Medicine and Therapeutics held on March 29th, 1997).
22. **Synthesis and Behavioral Study of Carboline Derivatives;** Z. S. Saify, Waqar Hussain, Javed Farhad, Nousheen Mushtaq, Fozia Noor, Shamoona Takween Shamim Akhtar, Khalid M. Khan and Shaheen A. Hussain; Proceeding of 2nd International conference on Pharmaceutical Sciences, Karachi. Jan 1997

RESEARCH PAPERS

1. **Synthesis of 9-Aminoacridine Derivatives as Anti-Alzheimer Agents** Rabya Munawar · Nousheen Mushtaq · Sadia Arif · Ahsaan Ahmed · Shamim Akhtar · Sumaira Ansari · Sadia Meer · Zafar S Saify · Muhammad Arif, *American Journal of Alzheimer s Disease and Other Dementias*, 2015, 1-7
2. **Antimicrobial and antioxidant screening of N'-substituted sulphonyl and benzoyl derivatives of 4-Pyridine carboxylic acid hydrazide**, Sabahat Naeem · Shamim Akhtar · Nadia Asghar · Sikarder Khan Sherwani · Nousheen Mushtaq · Arfa Kamil · Shaista Zafar · Mohammad Arif · Zafar Saeed Saify 2015· *Pak J. Pharm Sci* 28(6):2129-2134
3. **2-(2'-pyridyl) benzimidazole analogs and their β -glucuronidase inhibitory activity** A. Kamil · S. Akhtar · S. Noureen · Z.S. Saify · S. Jahan · K.M. Khan · F. Rahim · M. Taha · N. Mushtaq · M. Arif · S. Perveen · M.I. Chaudhary *Journal of the Chemical Society of Pakistan*. 2015, 37 (4) 787-791
4. **Analgesic and anxiolytic evaluation of N-substituted arylsulphonyl and benzoyl derivatives of 4-Pyridine-carbohydrazide**. Sabahat Naeem, Shamim Akhtar, **Nousheen Mushtaq**, Arfa Kamil, Iffat Mehmood, Shaista Zafar, Muhammad Arif and Z. S. Saify. *FUUAST Journal of Biology* 2014; 4(2):123-126.
5. **Synthesis and Biological Screening of 4-(1-Pyrrolidinyl) Piperidine Derivatives As Effective Analgesics**. Zafar Saeed Saify, Shazia Haider, Nighat Sultana, **Nousheen Mushtaq**. And Tabinda Zarreen. *Journal of chem. society of Pakistan*. 2014 36(5).
6. **Synthesis and Pharmacological evaluation of novel Benzoyl derivatives of Piperidine-4-carboxamide**. Azra Rauf, Shamim Akhtar, Sabahat Naeem, **Nousheen Mushtaq** and Muhammad Arif. "*IJRPC* , 4(3), 2014 pp. 509-516
7. **Synthesis of novel derivatives of 4-Pyridine carboxylic acid hydrazide and their activity on central nervous system** . Sabahat Naeem, Shamim Akhtar, **Nousheen Mushtaq**, Arfa Kamil, Shaista Zafar, Nadia Asghar, Sana Anwar and Muhammad Arif. “. *Pak. J. Pharm. Sci.* 27(5) (special), 2014, pp. 1401-1408.
8. **Determination of pKa values of new phenacyl-piperidine derivatives by potentiometric titration method in aqueous medium at room temperature (25±0.5°C)**”Shaista Zafar, Shamim Akhtar, Talat Tariq, **NousheenMushtaq**, Ahsaan Ahmed, Muhammad Arif and Sana Anwar. “. *Pak. J.Pharm .Sci.* 27 (4) 2014 pp 925-929.
9. **Synthesis of 4-(1-pyrrolidinyl)piperidine derivatives as antibacterial and antifungal agents**. ShaziaHaider, Z.S. saify, **NousheenMushtaq**, Mehrun- Nisa, Tabinda Z., M. A. Arain, S. M. ghufan Saeed and Seema Ashraf. *World. j. Pharm. Res*, 3 (3) 3723-3730, 2014
10. **Design and synthesis of potent and selective beta glucuonidase inhibitors by virtual and in vitro screening** ShaziaHaider, Z.S. saify, Mehrun- Nisa, **NousheenMushtaq**, Afshan Naz, Ajmal Khan, Bishnu P. Marasani, seema Ashraf, Tabinda Z. and M. A. Arain, S. M. ghufan Saeed and Seema Ashraf. “”*World j. Pharm. Sci.*, 2(5) 455-461, 2014

11. **Synthesis and Cytotoxic activity of some derivatives of alkyl piperidine** Sarwat Jahan, Shamim Akhtar, Z. S. Saify, **Nousheen Mushtaq**, Ali Akber Sial, Arfa Kamil and Muhammed Arif. “”. *Pak. J. Pharm.Sci.* 26(3), 2013, pp. 517-523.
12. **Disposal Practices of unused and expired Pharmaceuticals in Karachi and their impact on health and environment** Ahsaan Ahmed , **Nousheen Mushtaq**, Shamim Akhtar, Muhammad Arif and Ghazala Yasmeen; JUMDC Vol. 4, Issue 2, July-December 2013.
13. **Investigation of Piperidine derivatives in Ex Vivo Models of pain and platelet aggregation** Zafar Saeed Saify, Huma Rasheed, Nousheen Mushtaq, Mehrun Nisa, Shazia Haider, Afshan Naz, Kaniz Fizza Azhar, and G.A. Miana, *Arch Pharm Res* 35 (11) 1953-59 2012
14. **Behavioral and Neurochemical Profile of some Novel Phenacyl based Isonipecotamide Derivatives** Shamim Akhtar; M. Arif; Nousheen Mushtaq; Z.S. Saify, Ahsaan Ahmed and D.J. Haleem. *Pak. J. P. S.* 25(4), 2012, 705-714.
15. **Characterization of Plasmodium falciparum aspartic protease inhibition by piperidine derivatives.** Zafar Saied Saify, Mehrun Nisa, Kaniz Fizza Azhar, M. Kamran Azim, Huma Rasheed, Nousheen Mushtaq, M. Arshad Arain, **Shazia Haider**, Manawar Khanum, and Waseem Ahmed (2011). *J. of Nat. Prod. Res.* 2011 Dec;25(20):1965-68
16. **Synthesis of some novel analogues of 4-(1-pyrrolidinyl) piperidine and their effect On Plasma glucose level** Nousheen Mushtaq , Z.S. Saify, Shamim Akhtar, Muhammad Arif, Saida Haider and Nazish Saba, *Pak. J. Pharm. Sci.* 2010 vol. 23 (2), 220-23
17. **Molecular approaches toward Alzheimer’s disease.** Faiza Tahir, Att-ur-Rahman, M. Iqbal Choudry, Z.S. Saify, Mehrunnisa, Shazia Haider and Nousheen Mushtaq; *Alzheimer’s and Dementia*, 4(4), suppl 2008 ppT474
18. **Synthesis and Behavioral study of 4-(1-pyrrolidinyl)Piperidine and its derivatives** Nousheen Mushtaq, Z.S. Saify, Shamim Akhtar Viqar-uddin Ahmed and M.Arif *Pak. J. Pharmacol.* 2008 vol. 25 (1), 19-24
19. **Synthesis and Evaluation of novel phenacyl based carboxamide derivatives of Piperidine as antihypertensive agents** Shamim Akhtar, Z. S. Saify, M. Arif, Nousheen Mushtaq, Ahsana Dar and Ashfaq Ahmed. *Pakistan Journal of Physiology*, 2006 2 (1), 42-8
20. **Synthesis and biological activities of novel 4-(4'-chlorophenyl)-4- hydroxypiperidine derivatives** Z.S. Saify , Nousheen Mushtaq, Khalid M. Khan, S. Perveen , S.T. Shah, Abdel-Jalil RJ, M. Fecker, Voelter W. *Chem Pharm Bull* 2005 Jan;53(1):64-6.
21. **Antibacterial Activity of 1-methyl-7-methoxy-β-carboline and its phenacyl and coumarine analogues** Z.S. Saify, Jawaid Farhad, Nousheen Mushtaq, Fozia Noor, Shamim Akhtar Baqir Shyum Naqvi, Muhammad Haris Shoaib and Khalid M. Khan *Pak. J. Pharm. Sci*, 2005 vol.18 (3), 39-41
22. **Neurochemical estimations of some new quaternary phenacyl-bromo-piperidinium compounds** Z.S. Saify, Shamim Akhtar ,Muhammad Arif, Nousheen Mushtaq and D.J. Haleem, *Pak. J. Pharm. Sci*, 2005; vol. 18 (2), 52-4

23. **Effects of Fish oil and lipid lowering drugs on serum triglycerides alone and in combination.** Z.S. Saify, Faheem Ahmed, Shamim Akhtar, Nousheen Mushtaq, M. Arif and Sonia Siddiqui. *Pak. J. Pharmacol.* 2004, Vol 21(2), 1-7.
24. **Effect of Diazepam on CNS excitation and other behavioral changes induced by Harmaline and its newly synthesized analogues;** Z. S. Saify, Jawaaid Farhad, Nousheen Mushtaq, Fozia Noor, Shamoona Takween, Shamim Akhtar and Khalid M. Khan; *Pak. J. Pharm. Sci.* 2004 vol. 17 (1),31-6
25. **Biochemical studies on Marine Fish oil Part-I: Effects of fish oil and lipid lowering drugs on LDL /HDL cholestrol levels** Z.S. Saify, Faheem Ahmed, Sonia Siddiqui, Shamim Akhtar, Sohail Hasan, Shaheen A. Hussain and Nousheen Mushtaq; *Pak. J. Pharm. Sci.* 2003 vol. 16 (2) 1-8.
26. **Synthesis of some phenacyl derivatives of 1-methyl-7-methoxy-beta carboline and their behavioral study** Z.S. Saify, Jawaaid Farhad, Nousheen Mushtaq, Fozia Noor, Shamoona Takween, Shamim Akhtar, Waqar Hussain, Shaheen A. Hussain and Khalid M. Khan, *Pak. J. Pharm. Sci.* 2003.Vol. 16 (1) 73-7
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